Or Hadas

hadas.or11@gmail.com   Website
Linkedin   Scholar

Education

2020 – 2026	**Ph.D., Weizmann Institute of Science** Atmospheric Dynamics Thesis title: *From Individual Storms to Climate: Bridging Lagrangian and Eulerian Perspectives on Midlatitude Atmospheric Dynamics*
2017 – 2020	**M.Sc., Weizmann Institute of Science** Atmospheric Dynamics Thesis title: *Suppression of baroclinic eddies by strong jets.*
2016 – 2017	**Aviation Meteorology** Israeli Air Force.
2011 – 2016	**B.Sc., Tel Aviv University** in Physics. Started at the age of 14.

Employment

2017 – 2026	**Researcher,** Yohai Kaspi's Atmospheric Dynamics Lab, Weizmann Institute of Science. Data analysis and modeling of the atmosphere as part of M.Sc. and Ph.D. work.
2020 – 2023	**Teaching Assistant,** The Weizmann School of Science. Provided tutoring for *Atmospheric and Oceanic Fluid Dynamics* and *Climate Change Debates*.
2017 – 2019	**Meteorologist,** Israeli Air Force.
2015 – 2016	**Researcher,** Yoram Dagan's Condensed Matter Lab, Tel Aviv University. As part of the program for excellent undergraduate physics students.

Research Interests

- Climate Dynamics and Modeling, Parameterization, Machine Learning and Information Theory in Climate Science

Methods & Tools

Numerical Analysis	Using Matlab, Numpy, Scipy, Sklearn (8 years)
Parallel computing	Using clusters, utilizing various scheduling software (PBS, SLURM, etc., 6 years).
Big Data	Analysis of large datasets (ERA-5, CMIP6, etc.) using Dask (6 years)
Climate Modeling	Running and analyzing data from the FMS and CESM GCMs (8 years)
Machine Learning	Building and training models using Tensorflow (3 years)

Publications

1. **O. Hadas** and Y. Kaspi, "Quantifying the influence of climate on storm activity using machine learning," *Geophys. Res. Lett.*, vol. 53, e2025GL118496, 2026. DOI: 10.1029/2025GL118496.
2. **O. Hadas** and Y. Kaspi, "Stronger jet, weaker storms: A mechanistic perspective on the atlantic-pacific storm paradox," *Nat. Commun.*, 2026.

3 W. Yao*, **O. Hadas***, and Y. Kaspi, “Predictability of storms in an idealized climate revealed by machine learning,” *Geophys. Res. Lett.*, vol. 53, e2025GL118886, 2026, *Equal contribution; supervised the research. DOI: 10.1029/2025GL118886.

4 **O. Hadas** and Y. Kaspi, “A Lagrangian perspective on the growth of midlatitude storms,” *AGU Adv.*, vol. 6, no. 3, e2024AV001555, 2025. DOI: 10.1029/2024AV001555.

5 J. E. Blanco, R. Caballero, G. Datseris, *et al.*, “A cloud-controlling factor perspective on the hemispheric asymmetry of extratropical cloud albedo,” *J. Climate*, vol. 36, no. 6, pp. 1793–1804, 2023. DOI: 10.1175/JCLI-D-22-0410.1.

6 **O. Hadas**, G. Datseris, J. Blanco, *et al.*, “The role of baroclinic activity in controlling Earth’s albedo in the present and future climates,” *P. Natl. Acad. Sci.*, vol. 120, no. 5, e2208778120, 2023. DOI: 10.1073/pnas.2208778120.

7 G. Datseris, J. Blanco, **O. Hadas**, *et al.*, “Minimal recipes for global cloudiness,” *Geophys. Res. Lett.*, vol. 49, no. 20, e2022GL099678, 2022. DOI: 10.1029/2022GL099678.

8 **O. Hadas** and Y. Kaspi, “Suppression of baroclinic eddies by strong jets,” *J. Atmos. Sci.*, vol. 78, no. 8, pp. 2445–2457, 2021. DOI: 10.1175/JAS-D-20-0289.1.

9 T. Tamarin-Brodsky and **O. Hadas***, “The asymmetry of vertical velocity in current and future climate,” *Geophys. Res. Lett.*, vol. 46, no. 1, pp. 374–382, 2019, *Equal contributors. DOI: 10.1029/2018GL080363.

Fellowships and Awards

2026 **Zuckerman STEM Leadership Programs**, Zuckerman Family Foundation link

Rotchild Postdoctoral Fellowship, Yad Hanadiv Foundation link

2025 **Harry H. Hess Postdoctoral Fellowship**, Department of Geosciences, Princeton University link

Editor’s Highlight on Eos.org (top 2% of papers), American Geophysical Union link

2023 **Grant to support participation in the WCRP Conference**, World Meteorological Organization

2022 **Pearlman Prize for Student-Initiated Research**, Weizmann Institute of Science

Azrieli Graduate Fellows Program, Azrieli Foundation link

2021 **30 Under 30 list**, Forbes Israel link

Graduate Studies Fellowship, IES, Weizmann Institute of Science link

2020 **Dean’s Prize for Outstanding Master’s Graduates**, Weizmann Institute of Science

2018 **Award of Excellence**, Meteorological Unit, Israeli Air Force

2015 **Research Program for Outstanding Students**, Physics Department, Tel Aviv University link

2011 **Presidential Program for Scientists of the Future** link

Outreach & Service

2022–2026 **Peer Reviewer for Academic Journals.**
Journal of the Atmospheric Sciences; Journal of Climate; Quarterly Journal of the Royal Meteorological Society; Weather and Climate Dynamics; Nature (co-reviewer)

2025 **Guest Speaker,** Future Scientists Center.
Talk: *"The Silent Revolution of Weather and Climate Predictions"* link

2024 – 2025 **Tutor,** Odyssey Program
Mentored high school students in physics and math (as a program alumnus) link

2023 **Guest Speaker,** Tel Aviv Youth University.
Talk: *"Predicting the Unpredictable: Understanding Cloud Behavior"* (Wolf Prize Laureates youth event) link

Outreach & Service (continued)

2021 – 2022 **Editorial Assistant,** Shakuf.
Co-edited climate change and sustainability newsletter HaGahlilit. link

2020 **Outreach article,** Future Scientists Center.
From the Imagination, through the Pen, to the Machine: Forecasting and Understanding the Weather. link

2020 – 2021 **Reviewer and Content Writer,** ClimateScience.
Reviewed educational content and contributed outreach articles link

2019 **Guest Speaker,** Future Scientists Center.
Talk: *"The Elevation of Man over the Spirit (also 'Wind' in Hebrew): Weather Forecasting"*

International Oral Presentations

- Columbia University — SEAS Colloquium in Climate Science, New York, NY, February 2026, *From Individual Storms to Climate: Bridging Lagrangian and Eulerian Perspectives on Midlatitude Atmospheric Dynamics*
- Harvard University — EPS ClimTea Seminar, Cambridge, MA, February 2026, *From Individual Storms to Climate: Bridging Lagrangian and Eulerian Perspectives on Midlatitude Atmospheric Dynamics*
- Massachusetts Institute of Technology — EAPS Sack Lunch Seminar, Cambridge, MA, February 2026, *From Individual Storms to Climate: Bridging Lagrangian and Eulerian Perspectives on Midlatitude Atmospheric Dynamics*
- Princeton University — Department of Geosciences Atmospheric and Oceanic Sciences Seminar, Princeton, NJ, February 2026, *From Individual Storms to Climate: Bridging Lagrangian and Eulerian Perspectives on Midlatitude Atmospheric Dynamics*
- American Meteorological Society — 25th Conference on Atmospheric and Oceanic Fluid Dynamics, Austin, TX, January 2026, *Impact of Jet Stream Orientation on Northern Hemisphere Winter Storm Activity*
- American Geophysical Union — AGU Fall Meeting, New Orleans, LA, December 2025, *Jet Orientation Controls on Northern Hemisphere Winter Storm Intensities*
- Midlatitude Storm-Tracks Workshop, Rosendal, Norway, June 2025, *Pacific and Atlantic Winter Storms Differ Due to Jet Stream Orientation*
- European Geosciences Union — EGU General Assembly, Vienna, Austria, April 2025, *Quantifying the Internal Variability of Midlatitude Storms Using Deep Learning*
- American Meteorological Society — 24th Conference on Atmospheric and Oceanic Fluid Dynamics, Burlington, VT, June 2024, *A Lagrangian Perspective on Baroclinic Growth of Midlatitude Storms*
- European Geosciences Union — EGU General Assembly, Vienna, Austria, April 2024, *The Response of Synoptic-Scale Weather to Jet Stream Characteristics: a Lagrangian Perspective*
- World Climate Research Programme — WCRP Open Science Conference, Kigali, Rwanda, October 2023, *The Role of Baroclinic Activity in Shaping Earth's Albedo in Present and Future Climates*
- The Role of Atmospheric Dynamics for Climate and Extremes Workshop, Munich, Germany, October 2023, *The Lagrangian Response of Storms to Changes in Atmospheric Forcing*
- Japan–Israel Joint Symposium on Atmospheric Dynamics and Climate, Tokyo, Japan, September 2023, *Baroclinic Growth from a Lagrangian Perspective*
- American Geophysical Union — AGU Fall Meeting, San Francisco, CA, December 2022, *The Role of Baroclinic Activity in Shaping Earth's Albedo in Present and Future Climates*

- Rossbypalooza 2022: Clouds and Convection in Diverse Climates, Chicago, IL, July 2022, *Is Southern Hemisphere Stormier because it is cloudier or vice versa?*
- NOAA Geophysical Fluid Dynamics Laboratory — Seminar, Princeton, NJ, July 2022, *Interpreting Abnormal Phenomena in the Midlatitude Climate: the Midwinter Suppression and the Hemispheric Albedo Symmetry*
- American Meteorological Society — 23rd Conference on Atmospheric and Oceanic Fluid Dynamics, Breckenridge, CO, June 2022, *Suppression of Baroclinic Eddies by Strong Jets: A Mechanism for the Pacific Mid-Winter Minimum*
- Midlatitude Storm-Tracks Workshop, Saint-Pierre-d'Oléron, France, May 2022, *The Role of Baroclinic Activity in Shaping Earth's Albedo in Present and Future Climates*
- European Geosciences Union — EGU General Assembly, Vienna, Austria, May 2022, *The Role of Baroclinic Activity in Shaping Earth's Albedo in Present and Future Climates*
- American Geophysical Union — AGU Fall Meeting, New Orleans, LA, December 2021, *Suppression of Baroclinic Eddies by Strong Jets*